Choose Inputs and Outputs

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Helpful Classroom Sorter

TEACHER GUIDE · 1 OF 9

Choose Inputs and Outputs

> Prepare Choose Inputs and Outputs: check the target, prerequisites, resources and adaptable teaching schedule.

Teach the learning cycle

Primary target: Turn the project outcome into precise **input** and **output requirement**s

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Sections 1–6; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min

> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Sections 9 and 11–14. Give extra time to finish the full task set; use Section 15 when the week includes a coding lab.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Retrieval answer guidance

Find a Real Problem

Without opening the old lesson, explain one key idea from Find a Real Problem. Give an example and a mistake that the idea helps you avoid.

Look for: Describe a specific user difficulty using supplied **evidence** without inventing its cause

Reference example, not the only acceptable wording: For example: fictional pupils choosing activities in the supplied classroom exercise had to choose again in four of ten requests because the first choice did not fit the time or room conditions.

Build an AI app names a solution or construction task without specifying a user difficulty or **evidence** of need.

Verify Important Outputs

Without opening the old lesson, explain one key idea from Verify Important **Output**s. Give an example and a mistake that the idea helps you avoid.

Look for: Separate an answer into claims and connect each claim to relevant **evidence**

Reference example, not the only acceptable wording: A directly stated fact is four rolls, 2 m per roll or twelve 50 cm pieces.

A derived quantity is 8 m starting length, 6 m used or 2 m remaining, supported by the relevant calculation from R. R contains no enjoyment **evidence**, so citing it does not support everyone enjoying the activity.

Privacy by Choosing Less

Without opening the old lesson, explain one key idea from Privacy by Choosing Less. Give an example and a mistake that the idea helps you avoid.

Look for: Justify which information a stated task needs and remove unnecessary fields

Reference example, not the only acceptable wording: Keep topic choice; remove name, full date of birth, address and photograph. The totals do not require identity or contact details.

Collecting extra fields creates copies and exposure to manage even if those fields are later ignored or deleted.

Use the student module and workbook section with this exact week title. The worked case, independent question and answers below form one aligned lesson sequence.

What the learner must demonstrate

> Design a paper/plastic sorter that receives an object description. Include a mixed paper-and-plastic cup. Specify **input**s, **output**s and the uncertain-case action. Is the owner's name useful?

Concept to explain

A useful design makes required information and promised result explicit.

Term	Meaning
required input	information necessary for the process
optional input	information that may help but is not required
output contract	specific result the system promises to produce
missing-data behavior	what happens when an input is absent

Prepare the class

Use the supplied scenario, student illustrations and blank paper. For numeric work, offer counters or a calculator after learners show the setup. Fictional records are supplied; personal data and paid accounts are unnecessary.

Alternative compact allocation

> This compact option is not the full-week schedule. Use the main teaching cycle and allow extra sessions for the investigation, independent work and project.

Allow 80 minutes: recall 8; explanation 12; worked demonstration 15; guided practice 20; independent assessment 15; project and reflection 10. Extend the lab into a second session when needed.

Preparation for the connected explanation

Use this catalogue and task contract unchanged through the shared project unless a later activity explicitly introduces a versioned change.

The **baseline** is deliberately complete and credible: careful manual checking can satisfy the specification.

The educational aim is a traceable, tested recommender and a defensible comparison, not a compulsory claim that automation wins. Minutes are exact paper-model values and resources are assumed available.

Readiness answer

It may miss **compatible** options on every request where one exists. Check useful suggestions and whether no-match results are justified, as well as in**compatible** suggestions.

The four-row B1 to B4 catalogue and fictional request cards

A **baseline** sheet with **input**, **output**, reason and pass or fail columns

Separate storage for teacher-reserved final cases, kept out of development work

End goal for the pictorial case

By the end, I can specify valid **input**s, exact **output**s and a manual **baseline**.

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

TEACHER GUIDE · 2 OF 9

Explain the mechanism

> Model Choose Inputs and Outputs using the supplied case, labelled mechanism and reasoning. Make the scope of each conclusion explicit.

Follow the information instead of imagining magic inside the machine.

1. Input — information entering

INPUT minimal required information

2. Process — the deciding step

PROCESS candidate decision **process**

3. Output — the result

OUTPUT specific promised result and uncertainty path

Key idea

CORE IDEA A useful design makes required information and promised result explicit.

How the mechanism works

UNDER THE HOOD Interface contracts define **input**s, **output**s, types, allowed values, **missing-data behavior**, and guarantees.

Good design avoids collecting data that cannot improve the promised result and avoids promising outcomes the **process** cannot support.

Draw a fresh example with all three stages:

Every step connects a claim to observable **evidence**.

The case to investigate

CASE A sorter can use item type and location found to suggest a bin without collecting a student's name or face.

Step 1

STEP 1 — OBSERVE minimal required information

Step 2

STEP 2 — TRACE candidate decision **process**

Step 3

STEP 3 — CONCLUDE specific promised result and uncertainty path

Model answer

MODEL REASONING Prefer the design that meets the measurable need with fewer, clearer **input**s and an honest **output**. State how missing values and ambiguous cases are handled before building.

A claim to avoid

DO NOT OVERCLAIM Every collected field creates cost and risk; every promised **output** needs **evidence**.

What new **evidence** could change this answer?

Concrete teaching case

The teaching case

A sorter suggests paper or plastic trays using material **observation**s.

The teaching case

Name an **input**, **output** and unnecessary **input**.

Exact explanation to model

Why the result follows

Input: material or a suitable image. **Output**: tray suggestion. Home address is unnecessary. Uncertain material needs review.

Explain the added worked example

Explain the added worked example · 1

Last week we identified a difficulty: some fictional activity choices did not fit the available time or room conditions. Now define the task precisely enough that two reviewers can agree whether an **output** is correct.

A **requirement** such as “suggest a good activity” is too vague. State which **input**s are accepted, what counts as **compatible** and what to do when no option fits.

Explain the added worked example · 2

For this paper project, a valid request contains an available time from 0 to 30 whole minutes and a quiet-room **requirement** written Yes or No. Yes means the activity must be marked quiet.

No means either quiet or non-quiet activities are permitted; it does not mean noise is required. A missing or invalid value needs clarification, not a guessed value or a no-match claim.

Explain the added worked example · 3

A **compatible** activity takes no more than the available minutes and meets the quiet-room condition. If several activities fit, suggest the first in catalogue order. If none fits a valid request, report no match.

Give a brief reason. This is a suggestion only: it does not book resources, tell anyone to begin or guarantee enjoyment.

Explain the added worked example · 4

These four rows are the complete approved catalogue for the paper model. Treat their times as fixed task values, not measured guarantees about real pupils. All required materials are assumed available.

The catalogue order B1, B2, B3, B4 is the **tie rule**; it is not a claim that B1 is everyone’s favourite.

Explain the added worked example · 5

For 10 minutes with quiet required, B1 fits because 10 is no more than 10 and B1 is quiet. B2 fails the quiet **requirement**. B3 and B4 take too long. For 8 minutes with quiet not required, B2 fits; B1 takes too long.

A request for 7 minutes has no **compatible** activity in this catalogue, even when quiet is not required.

Check each condition. A fast activity can still fail the quiet **requirement**. A quiet activity can still take too long. Exact equality is allowed: an activity listed as 10 minutes fits 10 available minutes.

A blank time is different from a valid time of zero.

Key terms

Term	Meaning
Requirement	A precise condition that the solution must satisfy
Baseline	A defined starting method used for comparison
Compatible	Meeting all the stated conditions of the request
Tie rule	A stated way to choose when several options qualify
Acceptance check	A check of whether a result satisfies a requirement

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

The manual checklist baseline

STAGE	MANUAL ACTION	RECORDED RESULT
Validate the request	Check minutes and Yes or No	Clarify invalid input
Inspect catalogue in order	Check time and quiet for each	Mark compatible rows
Choose or explain absence	First compatible row or none	Suggestion or no match
Keep the evidence	Record input, result, reason	Comparable baseline record

The baseline is a complete manual method. A prototype must be compared with this defined procedure, not an invented weak opponent.

Model the reasoning aloud · 6

Write the request at the top of a sheet. If it is invalid, record what needs clarification and stop. Otherwise mark each catalogue row as **compatible** or in**compatible** using both conditions.

Circle the first **compatible** row in order, or write no match if every row fails. Record the **output** and its reason. This checklist is our **baseline**.

Model the reasoning aloud · 7

The **baseline** can be correct on all cases when applied carefully. A prototype that matches it has not demonstrated better compatibility.

It may still be easier to use or faster to trace, but those are separate claims that need their own measures. Do not make the **baseline** skip a condition just to make the prototype look better.

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Choose Inputs and Outputs

A fictional classroom chooses a storage bin for model parts.

The facts we will use

> **Supplied facts**
>
> Catalogue order: B1 capacity 5, B2 capacity 9. Valid request is a whole part count 1–12.

> **Supplied facts**
>
> Return first bin with capacity at least the count; none if neither fits; clarify invalid **input**.

AI ACADEMY / AI-1/30

Choose Inputs and Outputs

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Validate request

Count 6 is whole and within 1-12.



2. Check first bin

B1 capacity 5 is below 6; reject it.



3. Check second bin

B2 capacity 9 is at least 6; accept it.



4. Baseline output

Report B2 and stop; save the comparison evidence.

CHANGE ONE THING

Change request 6 to 5.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Validate request	Count 6 is whole and within 1–12.
2. Check first bin	B1 capacity 5 is below 6; reject it.
3. Check second bin	B2 capacity 9 is at least 6; accept it.
4. Baseline output	Report B2 and stop; save the comparison evidence.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

The **baseline** is an ordered catalogue scan, not a preference ranking. It makes expectations testable before a prototype is built.

Count 10 is valid but has no matching bin; that differs from a blank count, which fails the **input** contract. Both situations need explicit **output** behavior.

What this example does not establish

Keep the stated limit

Bin capacity is the only supplied **fitting** criterion; weight or shape suitability cannot be inferred.

Change one thing in the pictured case

Change one thing

Change: Change request 6 to 5.

Predicted result

Prediction: The **baseline** returns B1.

Why the result follows

Reason: Its capacity equals the request, and at least includes equality.

Ask while showing the picture

Ask while showing the picture · 8

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

Demonstrate and guide practice

> Demonstrate Choose Inputs and Outputs, reduce support deliberately and check individual reasoning before discussing answers.

Board sequence

1. Put the facts on the board

Write the given facts: A sorter suggests paper or plastic trays using material **observation**s.

2. State the decision to explain

Write the question to explain: Name an **input**, **output** and unnecessary **input**.

3. Reveal the worked reasoning

Reveal after **prediction**s: **Input**: material or a suitable image. **Output**: tray suggestion. Home address is unnecessary. Uncertain material needs review.

Hold back the independent task

Keep for an independent attempt: A paper sorter must suggest pencil tray or book shelf. Choose one needed **input** and state the **output**. What should happen if the picture is blocked?

Classroom dialogue

Ask the class

Ask: Which facts are given, and which would we have to assume? Learners mark relevant

details before discussing answers.

Language to model

Say: Predict first. Show the rule, information path or calculation that would produce the result. A correct guess is not yet an explanation.

Reveal reasoning one step at a time. Explain why a tempting alternative fails. Write units and denominators for arithmetic, trace conditions for decisions, and identify supporting sources for claims.

In pairs, one learner solves while the other checks **evidence**; exchange roles. Remove the worked answer before independent assessment.

Model these guided questions

12 Yes: B1 and B3 fit; B2 and B4 are non-quiet, and B4 is also too long. Choose B1 because it is first in catalogue order. 8 Yes: B1 and B3 are too long; B2 is not quiet; B4 fails both. Report no match.

Blank time requires clarification, not a guessed number. Exit: equal successful **output**s on five cases do not show better accuracy for the app.

A separate claim such as faster completion needs recorded times under comparable conditions, with errors and help also recorded.

Case-specific teaching sequence

Stage	Teacher action
Session 1 start 0 to 7 minutes	Retrieve why zero incompatible suggestions is not sufficient evidence of help.
7 to 20 minutes	Define valid input and both compatibility conditions; complete W1.
20 to 33 minutes	Read the complete catalogue and run W2 by hand.
33 to 45 minutes	Complete W3 and distinguish blank input from a valid zero.
Session 2 start 0 to 8 minutes	Retrieve the tie rule and the meaning of Quiet required No.

Stage	Teacher action
8 to 23 minutes	Complete the guided baseline and discuss a fair comparison plan.
23 to 38 minutes	Collect W4 independently, requiring reasons for each room result.
38 to 45 minutes	Complete W5 and the exit; save the contract and baseline for Week 31.

Reduce help deliberately

Completion practice

Completion task: Using the catalogue, record the **baseline output** and reason for 10 minutes with quiet required, 8 minutes with quiet not required, and 7 minutes with quiet not required.

Identify which no-match result is justified and why.

Reasoning to listen for

Teacher check: 10 Yes gives B1: 10 fits exactly and it is quiet. 8 No gives B2: B1 is too long, B2 fits and non-quiet is permitted. 7 No gives no match: even the shortest activity requires 8 minutes.

That no-match result is justified by the complete catalogue.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual reasoning check

Use the worked case for **Requirement**s and a simple **baseline**. Proposed explanation: “Quiet required No excludes quiet activities”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Expected repair

Expected correction: No removes the quiet restriction; it does not reverse it.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictorial practice question

Trace requests 10 and blank. Why should they not receive the same explanation?

Reasoned teacher answer

10 is valid but neither bin fits: no match. Blank is missing **input**: request clarification before searching.

Changed-case reasoning: The **baseline** returns B1. Its capacity equals the request, and at least includes equality.

Collect each learner's explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Run the weekly evidence lab

> Run the Choose Inputs and Outputs investigation and use the first unsupported step to choose a teaching response.

Predict first, test honestly, and explain what the **evidence** means.

1. Predict

State your **prediction** before the result is known.

2. Investigate

Complete the task: Sketch two designs and remove an unnecessary **input**.

3. Compare cases

Keep a normal case and an edge case clearly separated.

4. Record honestly

Record what happened—even when it does not match your **prediction**.

5. Explain and plan the next test

Explain the result, one limitation, and the next useful test.

Use a fresh case after support

INDEPENDENCE After support is removed, repeat the proof with a fresh case.

Make the lab runnable

Use the concrete case as the shared normal case and the independent question as a second case. Each pair creates a boundary or missing-information case and predicts its result before testing.

Keep the expected result separate from an observed mistake.

Record	What to capture
Input or evidence	Exact data, instruction or claim
Expected result	Answer or safe fallback declared before the run
Observed result	Actual result, including errors
Explanation	Rule, calculation or evidence explaining the outcome
Next test	One justified change and a new prediction

Diagnose and reteach

> **Find the unsupported claim**
>
> CLAIM TO REPAIR A system can safely infer every missing **input**.

For guessing, return to given facts and request a trace. For vocabulary without application, use one example and one non-example. For arithmetic errors, check the setup before the calculation.

Finish with a different uncoached case before recording mastery.

Address these additional misconceptions

Quiet required No excludes quiet activities: No removes the quiet restriction; it does not reverse it.

Ten minutes does not fit a ten-minute limit: No more than includes equality.

Blank time means zero: Blank is missing; zero is a valid **input** with no matching activity here.

A **baseline** must be bad to make a comparison useful: Use a credible simple method and accept a tie or a **baseline** win.

Match the support to the first error

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

Assess and explain the answers

> Use the Choose Inputs and Outputs questions, teacher keys and criteria after the relevant attempts. Previously discussed cases are practice; protect the separately selected independent task.

Worked case answer

Teacher key · Worked case answer

Input: material or a suitable image. **Output**: tray suggestion. Home address is unnecessary. Uncertain material needs review.

Independent question

Independent question · withhold the key

A paper sorter must suggest pencil tray or book shelf. Choose one needed **input** and state the **output**. What should happen if the picture is blocked?

Expected independent answer

Teacher key · Expected independent answer

A clear object picture or relevant shape description can be an **input**; the suggested location is the **output**. A blocked picture should trigger a request for a clearer view or review.

What counts as evidence

Criterion	0 points	1 point	2 points
Result	Incorrect or absent	Partly correct	Correct with necessary conditions
Reasoning	No supporting trace	Some steps	Clear mechanism, calculation or source support
Limit	Overclaims certainty	Vague limitation	Relevant uncertainty or missing fact
Transfer	Copies without adapting	Adapts with prompts	Solves a changed case independently

Use 6/8 as a classroom checkpoint, with 2 required for both Result and Reasoning. This is a proposed teaching rule, not a validated certification threshold.

Do not count a coached answer as independent mastery; correct unsafe or unsupported claims regardless of total.

Feedback

Ask the learner to find the **evidence** that challenges this claim: “A system can safely infer every missing **input**.” Keep the first attempt and an uncoached retry separate.

Original transfer answer

Teacher key · Original transfer answer

Use material **evidence** as **input**; **output**s are paper, plastic or review. Send the mixed cup to review under a predefined local rule. Owner name does not determine material.

A description may be incomplete, so the sorter should not claim it has directly inspected the object.

Explanations for the additional workbook examples

W1 Make a requirement checkable

Workbook question · W1

Replace “suggest a good activity” with the time condition, quiet condition and **tie rule** used in this project. Explain why Quiet required No permits B1 as well as non-quiet activities when their times fit.

Teacher reasoning · W1

Time required must be no more than available time; when quiet is required the activity must be quiet. Choose the first **compatible** activity in B1 to B4 order.

No means quiet is not required, not forbidden, so quiet B1 is permitted if 10 minutes fits.

W2 Run the baseline on three requests

Workbook question · W2

Using the catalogue, record the **baseline output** and reason for 10 minutes with quiet required, 8 minutes with quiet not required, and 7 minutes with quiet not required. Identify which no-match result is justified and why.

Teacher reasoning · W2

10 Yes gives B1: 10 fits exactly and it is quiet. 8 No gives B2: B1 is too long, B2 fits and non-quiet is permitted. 7 No gives no match: even the shortest activity requires 8 minutes.

That no-match result is justified by the complete catalogue.

W3 Separate invalid input from no matching option

Workbook question · W3

Compare a request with blank minutes and Quiet required Yes to a request with 0 minutes and Quiet required Yes. What should the **baseline** record for each?

Explain why these are different outcomes even though neither produces an activity suggestion.

Teacher reasoning · W3

Blank minutes is invalid and requires clarification of the missing time. Zero is a valid whole number within 0 to 30; no activity takes 0 minutes or less, so no match is justified.

The first lacks a usable **input**; the second has a usable **input** but no **compatible** option.

W4 Build a baseline for a different catalogue

Workbook question · W4

A paper room recommender uses R1 first, capacity 6, quiet Yes; then R2, capacity 10, quiet No. Valid requests give a whole group size from 1 to 12 and Quiet required Yes or No.

A room fits when capacity is at least group size and the quiet condition is met; choose the first **fitting** room. Record **output**s and reasons for 6 people Yes, 8 people No and 8 people Yes.

Explain how a manual **baseline** can be compared fairly with a prototype. No real booking is allowed.

Teacher reasoning · W4

6 Yes gives R1: equality fits and it is quiet. 8 No gives R2: R1 is too small and R2 meets capacity with quiet not required. 8 Yes gives no match: R1 is too small, R2 is not quiet.

Use the same room catalogue, request set and **requirement**s for both methods and keep all results. Compare suggestions only; do not make bookings.

W5 Challenge an unfair comparison

Workbook question · W5

Team A tests a prototype on five easy requests and the manual **baseline** on five different requests, then reports only the prototype's successful **output**s. Identify two changes needed for a fair comparison.

If both methods then meet all **requirement**s, what can and cannot be concluded about compatibility?

Teacher reasoning · W5

Use the same requests and catalogue for both with the same judging rules, and retain all **output**s including failures and no-match or clarification results.

If both meet the **requirement**s on those cases, both pass those checks; the results do not demonstrate the prototype has better compatibility or prove either works on every possible case.

Assess independent transfer

Independent question · withhold the key

Selected case: A paper room recommender uses R1 first, capacity 6, quiet Yes; then R2, capacity 10, quiet No. Valid requests give a whole group size from 1 to 12 and Quiet required Yes or No.

A room fits when capacity is at least group size and the quiet condition is met; choose the first **fitting** room. Record **output**s and reasons for 6 people Yes, 8 people No and 8 people Yes.

Explain how a manual **baseline** can be compared fairly with a prototype. No real booking is allowed.

Selected case answer

Teacher key · Selected case answer

6 Yes gives R1: equality fits and it is quiet. 8 No gives R2: R1 is too small and R2 meets capacity with quiet not required. 8 Yes gives no match: R1 is too small, R2 is not quiet.

Use the same room catalogue, request set and **requirement**s for both methods and keep all results. Compare suggestions only; do not make bookings.

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Teacher key · Selected case answer

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Feedback that the learner uses

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

Connect and extend learning

> Connect Choose Inputs and Outputs to Helpful Classroom Sorter, adapt support and plan a later changed-case check.

Helpful Classroom Sorter

Freeze the sorter's v1 **input** list, **output contract**, and not-sure behavior.

Keep the decision, **input**, result, explanation and remaining limitation in the project **evidence** record. Retrieve this record next week rather than restarting the project.

Support and stretch

Use physical cards, spoken explanations and a three-stage drawing. Read aloud without giving the answer. Stretch: compare two **mechanism**s with the same visible **output** and explain what **evidence** distinguishes them.

Exit reflection

Ask for one decision the learner can now justify and one situation requiring more **evidence** or human review. Confidence ratings are separate from correct independent performance.

Bridge to the next week

Next: Week 31: Design the Decision **Process**. Retrieve today's **evidence** and identify what carries forward and what is new.

Support and fresh reassessment

Give one tick column for time or capacity and another for quiet. Require both ticks before circling a suggestion. Use a number line to distinguish equal to from greater than.

Extension: explain why a fixed **tie rule** helps reproducibility but does not prove a preference ranking.

W4 requires R1, R2 and no match with capacity and quiet reasons, plus matching **input**s, catalogue and judging rules in the comparison. After support change R2 to quiet Yes and repeat 8 Yes: expected R2.

Ask whether this revises a **requirement** or a catalogue value; it revises the catalogue, so record a new catalogue version and rerun affected checks.

Week 31 designs the complete decision **process** using the same **requirement**s and catalogue.

Plan the delayed return

Suggested return points in this level: ai-1/31, ai-1/33, ai-1/36. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Project evidence from this case

Document sorter schema, catalogue order, acceptance rule and all **output** types; use a manual trace as the comparison **baseline**.

Prepare, teach and assess

> Use this preparation summary alongside the detailed Choose Inputs and Outputs lesson; it is a checklist rather than a replacement for the full sequence.

Teaching target

> Teaching target: Turn the project outcome into precise **input** and **output requirement**s

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. No paid AI account is required for the paper reasoning activities.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
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> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
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> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Sections 9 and 11–14. Give extra time to finish the full task set; use Section 15 when the week includes a coding lab.

Before class, solve the supported practice: Using the catalogue, record the **baseline output** and reason for 10 minutes with quiet required, 8 minutes with quiet not required, and 7 minutes with quiet not required.

Identify which no-match result is justified and why.

Supported practice answer

Reasoned practice answer: 10 Yes gives B1: 10 fits exactly and it is quiet. 8 No gives B2: B1 is too long, B2 fits and non-quiet is permitted. 7 No gives no match: even the shortest activity requires 8 minutes.

That no-match result is justified by the complete catalogue.

Check every learner before discussion: Use the worked case for **Requirement**s and a simple **baseline**. Proposed explanation: “Quiet required No excludes quiet activities”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Repair the misconception

Misconception repair: No removes the quiet restriction; it does not reverse it.

Support without giving away the answer

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extend the reasoning

Extension: Change request 6 to 5. Require a **prediction** and a reason before comparing. Expected result: The **baseline** returns B1. Reason: Its capacity equals the request, and at least includes equality.

Protect the independent attempt

Independent assessment: withhold the reserved answer until the learner has attempted the

new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint

Project checkpoint: Prepare the demonstration. Gather reproducible instructions, **evidence**, access checks and remaining limitations.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

Use feedback, labs and repair paths

> Use feedback, earlier review and portfolios purposefully. Apply the coding routine only when this lesson has an executable lab.

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

For later coding lessons

> This week has no code block. Use this routine when a later lesson includes an executable lab.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

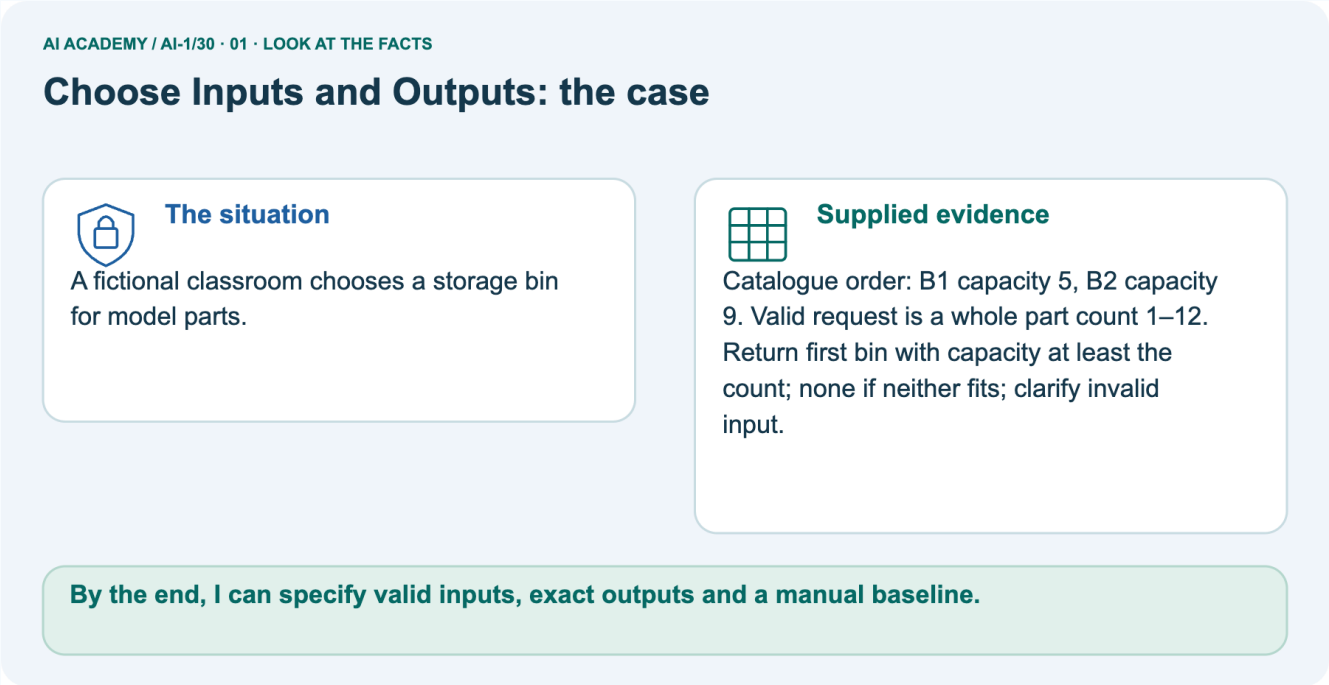
Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

Visual concept journey

> Show one Choose Inputs and Outputs graphic at a time, request a prediction and discuss its trace before the independent assessment.

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Choose Inputs and Outputs: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

A fictional classroom chooses a storage bin for model parts. Supplied facts: Catalogue order: B1 capacity 5, B2 capacity 9. Valid request is a whole part count 1–12.

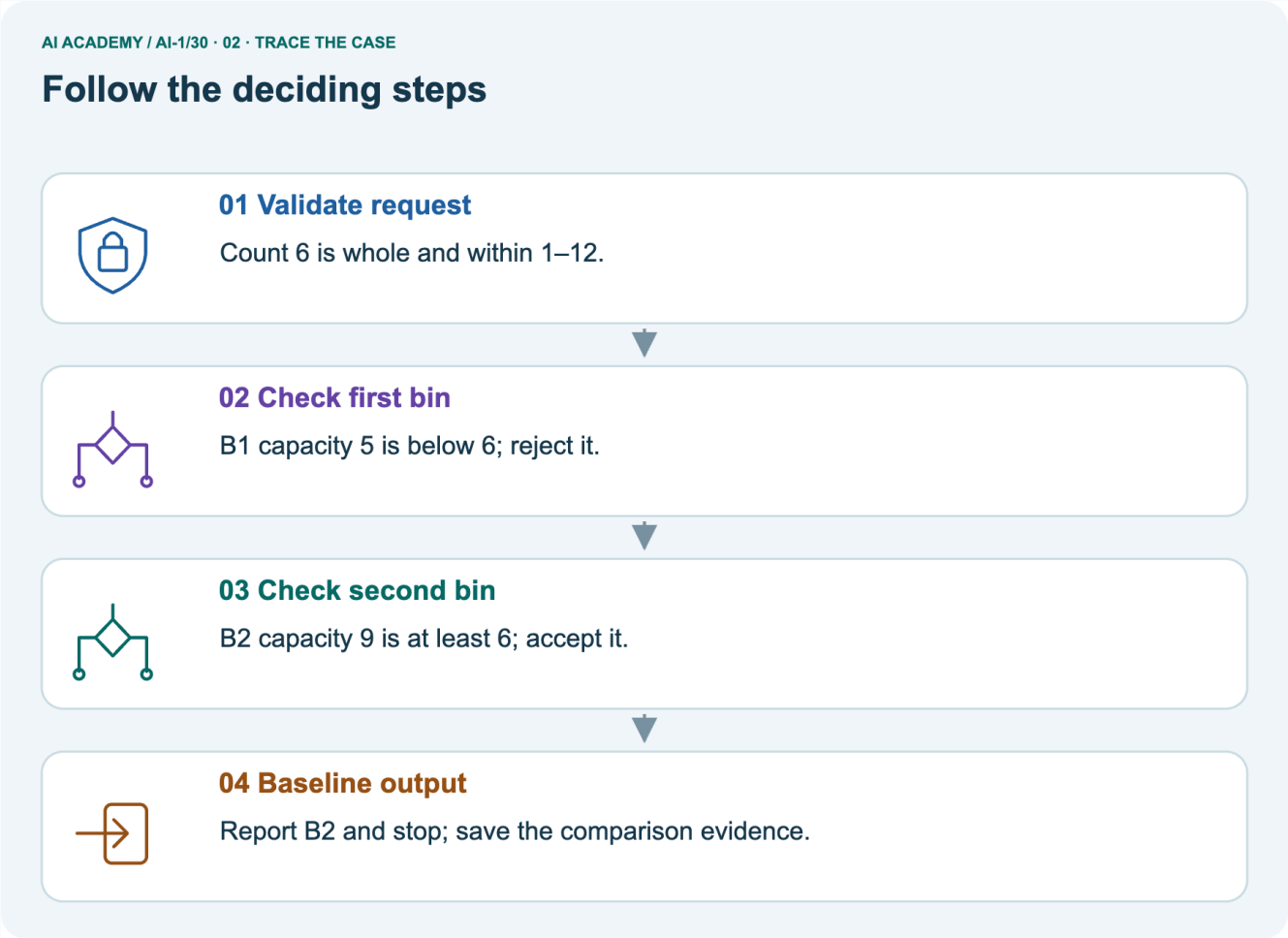
Return first bin with capacity at least the count; none if neither fits; clarify invalid **input**.

> Point to the given fact that matters most. Explain why it matters.

Discuss while showing the picture

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

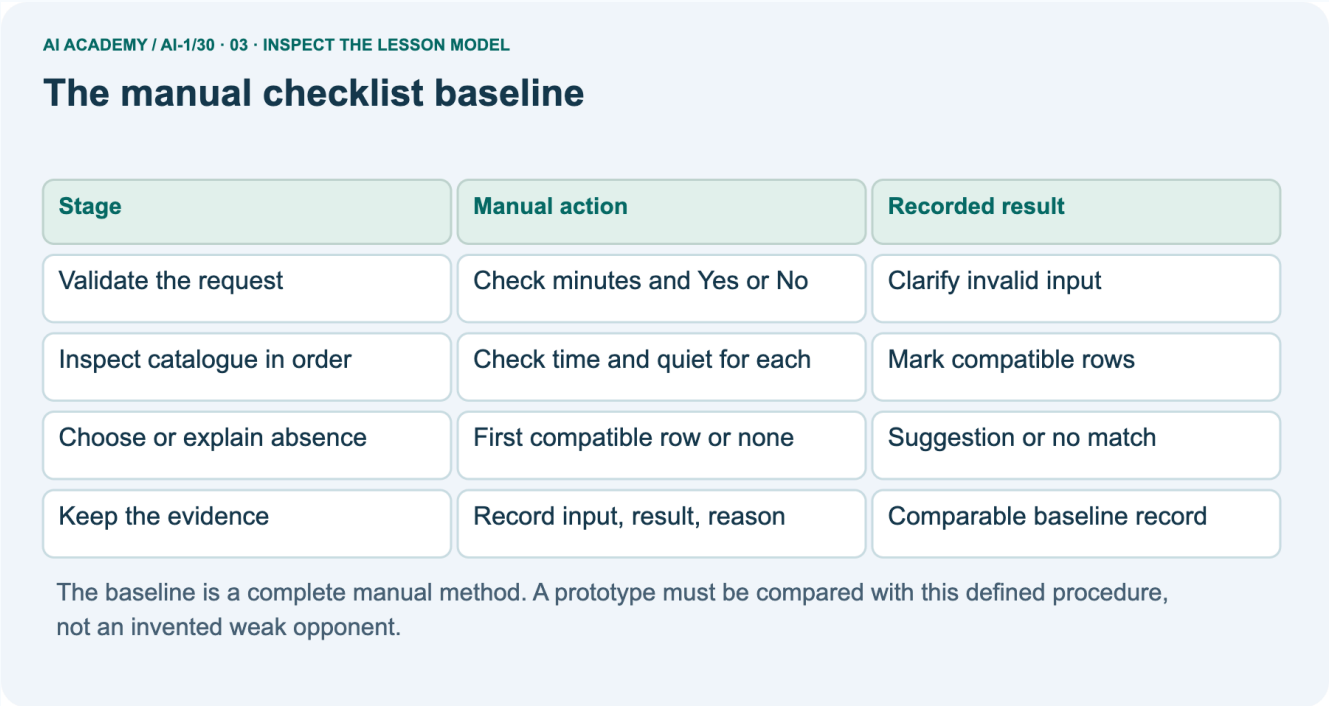
Validate request: Count 6 is whole and within 1–12. → Check first bin: B1 capacity 5 is below 6; reject it. → Check second bin: B2 capacity 9 is at least 6; accept it. → **Baseline output**: Report B2 and stop; save the comparison **evidence**.

> Point to each numbered stage and explain which recorded **input** it uses.

Discuss while showing the picture

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. The manual checklist baseline



Follow the labelled picture with the matching explanation. Open the vector diagram

The manual checklist **baseline**. Validate the request | Check minutes and Yes or No | Clarify invalid **input**. Inspect catalogue in order | Check time and quiet for each | Mark **compatible** rows.

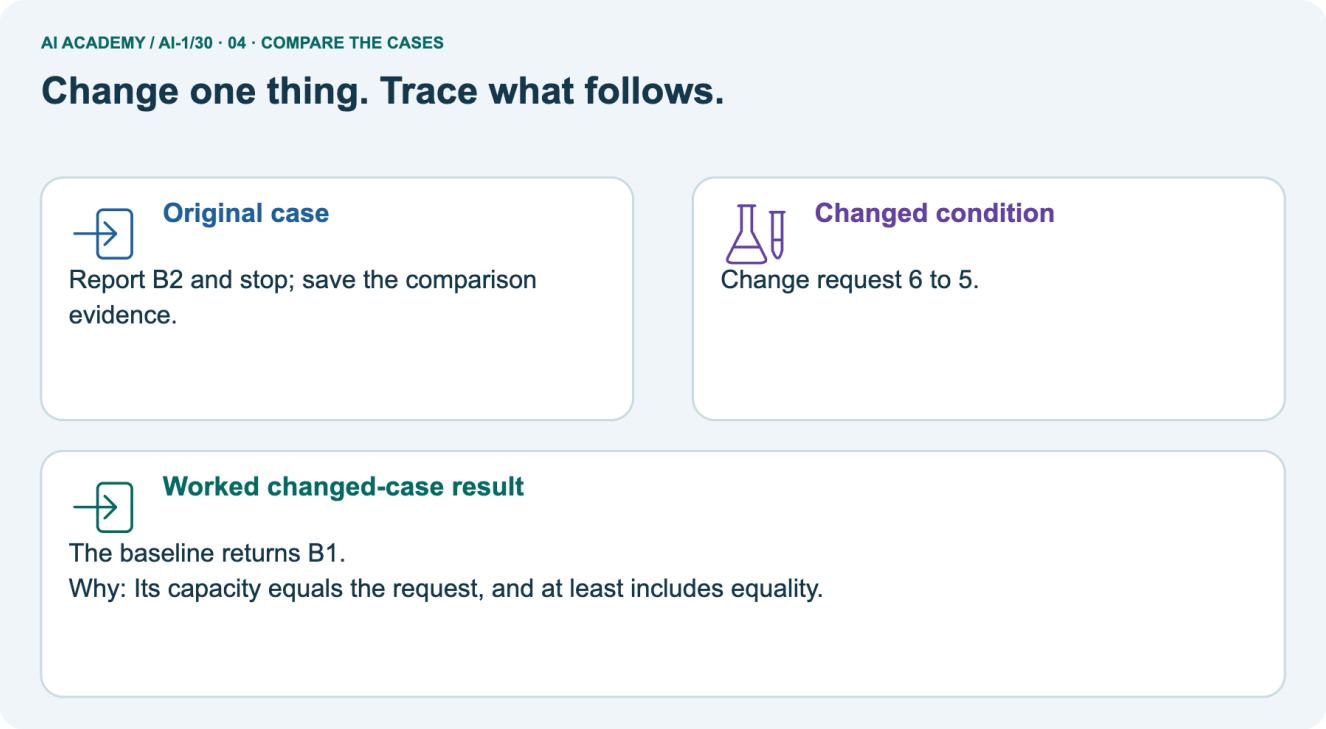
Choose or explain absence | First **compatible** row or none | Suggestion or no match. Keep the **evidence** | Record **input**, result, reason | Comparable **baseline** record

Read the labels and values. Explain the relationship the model shows.

Discuss while showing the picture

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Report B2 and stop; save the comparison **evidence**. Changed condition: Change request 6 to 5. Expected result: The **baseline** returns B1. Reason: Its capacity equals the request, and at least includes equality.

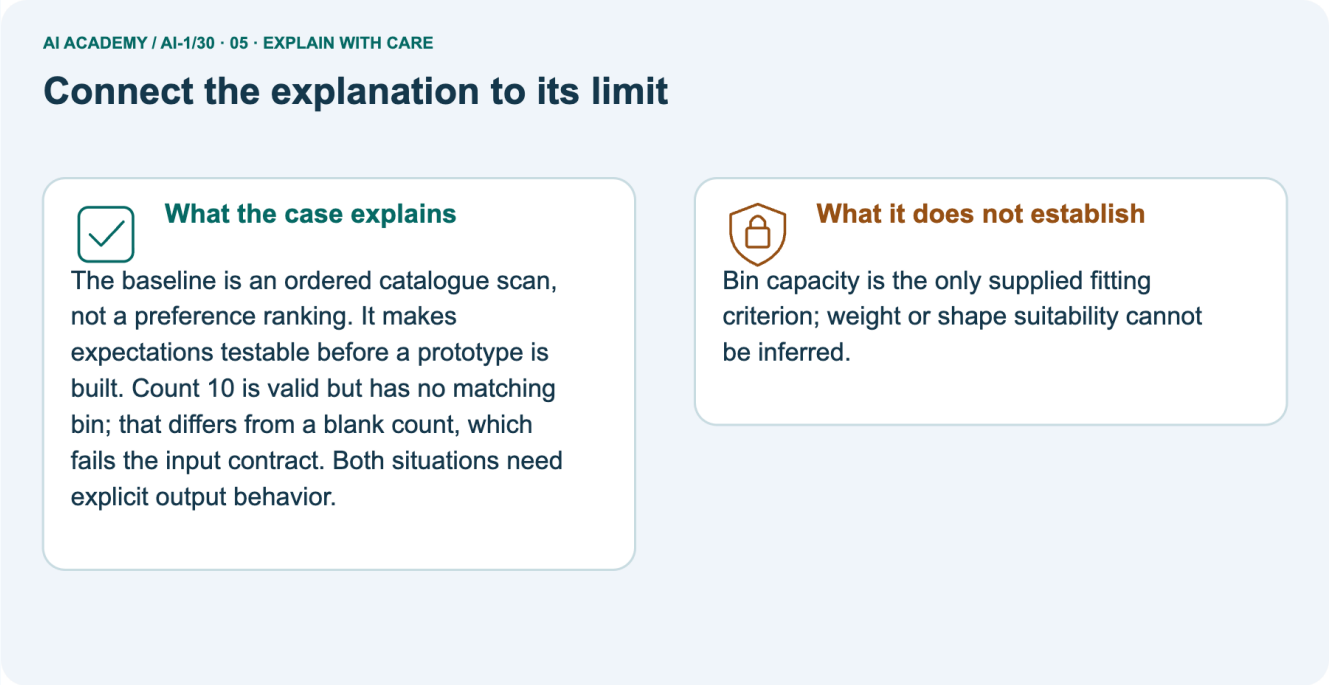
Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Discuss while showing the picture

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: Its capacity equals the request, and at least includes equality.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

The **baseline** is an ordered catalogue scan, not a preference ranking. It makes expectations testable before a prototype is built.

Count 10 is valid but has no matching bin; that differs from a blank count, which fails the **input** contract. Both situations need explicit **output** behavior.

Limit: Bin capacity is the only supplied **fitting** criterion; weight or shape suitability cannot be inferred.

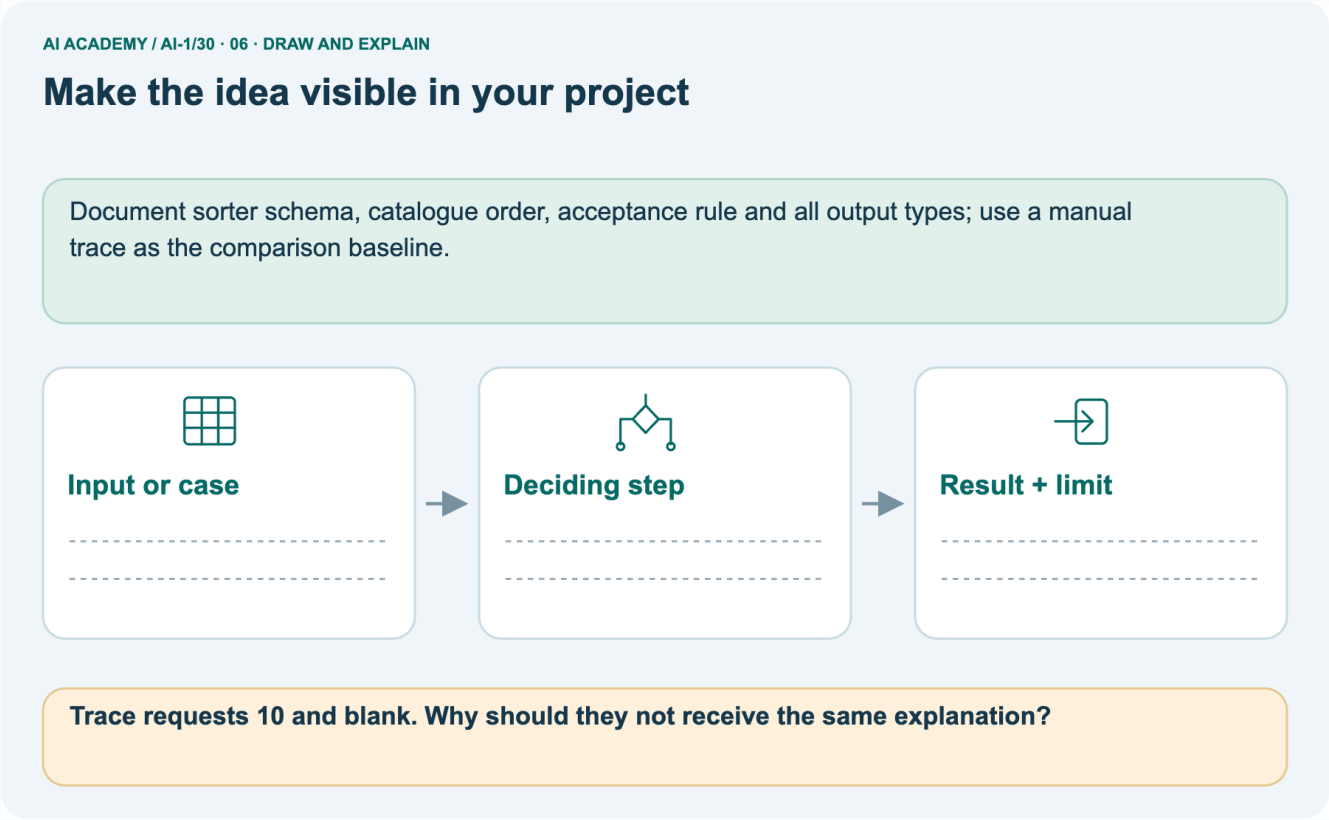
Explain the case, then point to a limit on the claim.

Discuss while showing the picture

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: Bin capacity is the only supplied **fitting** criterion; weight or shape suitability cannot be inferred.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Document sorter schema, catalogue order, acceptance rule and all **output** types; use a manual trace as the comparison **baseline**. Draw your **input**, deciding step, result and limit. Supported question: Trace requests 10 and blank.

Why should they not receive the same explanation?

Draw your project trace and justify the deciding step using **evidence**.

Discuss while showing the picture

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: 10 is valid but neither bin fits: no match. Blank is missing **input**: request clarification before searching.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.